

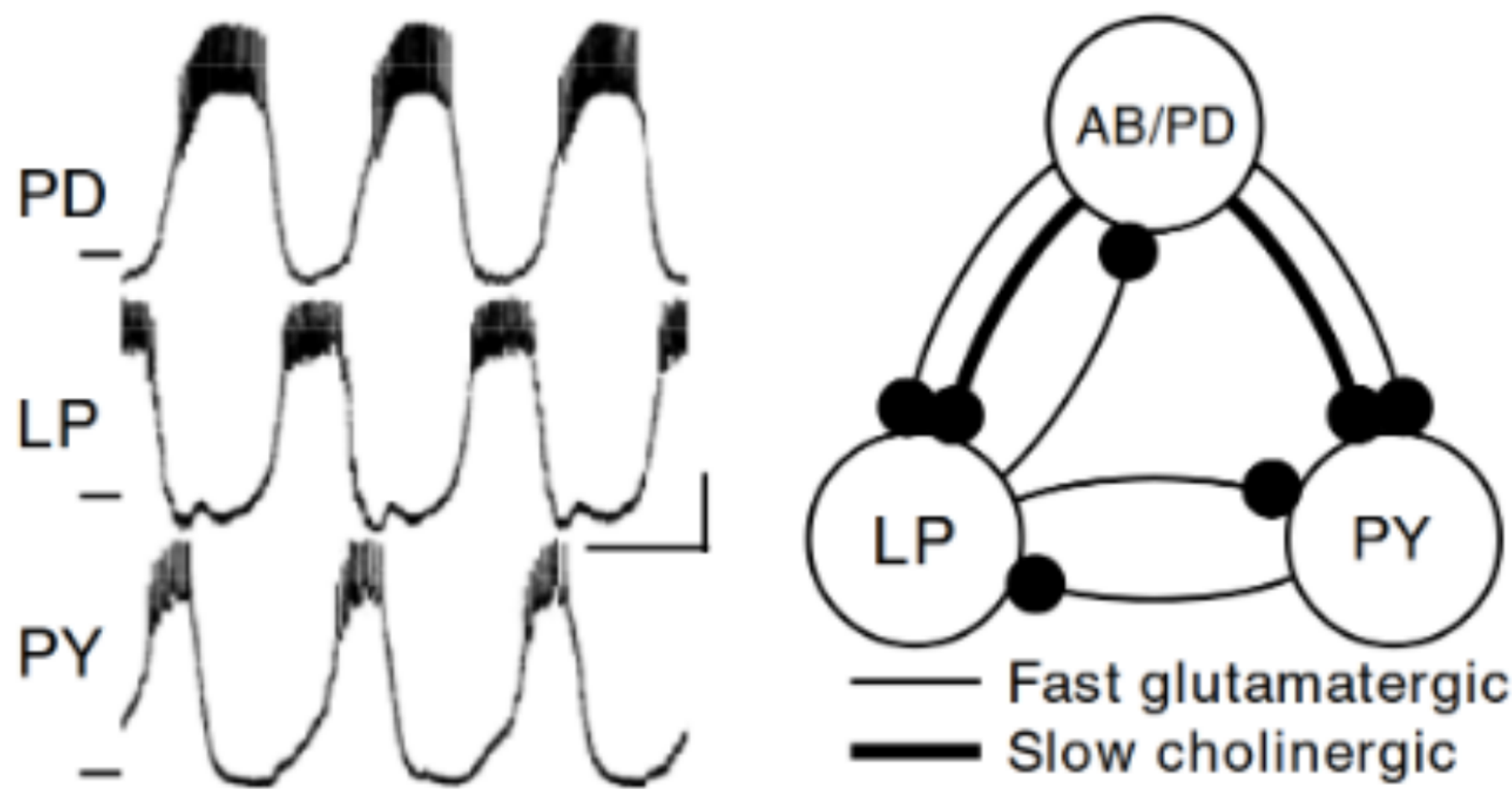
Motivation

The stomatogastric ganglion (STG) pyloric circuit produces a robust **triphasic rhythm** that drives feeding movements in crustaceans. A key question in computational neuroscience:

“What synaptic conductances gave rise to a 4-second voltage recording, and is this parameter set unique?”

Prinz et al. (2004) showed that many distinct parameter combinations produce functionally equivalent rhythms — **degeneracy**. We tackle both inference and degeneracy simultaneously with two methods.

Model: Simplified STG Circuit



A 3-neuron Jaxley model (AB/PD, LP, PY) with **7 free synaptic conductances**:

Synapse	Type
AB/PD → LP (×2)	Glut + Chol
AB/PD → PY (×2)	Glut + Chol
LP → AB/PD	Glutamatergic
LP → PY	Glutamatergic
PY → LP	Glutamatergic

Parameters are searched in **log-space**: $\theta_i = \log_{10}(g_i)$, prior $\theta_i \sim \mathcal{U}(-5, 1) \mu S$.

Summary Statistics (9 features)

#	Feature
0	Burst period (ms)
1–3	Duty cycle: AB/PD, LP, PY
4–5	Phase offset: LP, PY (re AB/PD)
6–8	Spikes/burst: AB/PD, LP, PY

Phase offsets are **weighted 3×** in the DE loss because they capture the triphasic coordination most sensitively. For non-bursting simulations, a NaN penalty of 10 is assigned.

Methods

Differential Evolution (DE)

Gradient-free global optimiser (SciPy). Treats the simulator as a black box — no differentiability required.

- Population size: 5, max 50 generations
- Loss: weighted fractional error on 9 features
- Best result: loss = 0.0073**

Why not gradient descent?

Spike counts and burst periods are **step functions** — gradients vanish. Raw voltage MSE is phase-sensitive and traps gradient descent in local minima (wrong phase, right period). DE sidesteps both problems.

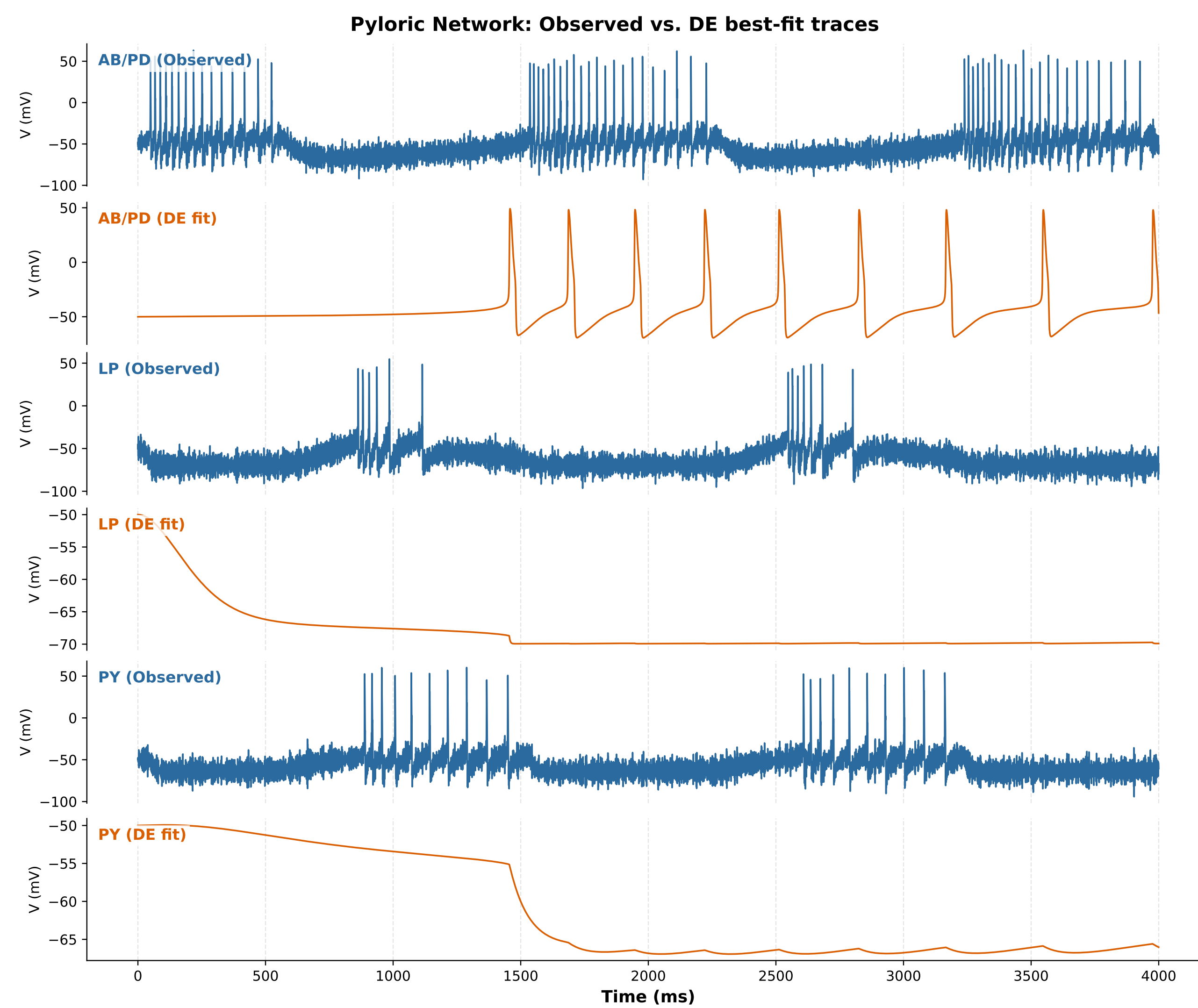
SNPE (Sequential NPE)

Amortised posterior inference using neural density estimation (sbi, SNPE-C / NPE).

- Prior: log-uniform over $\log_{10}(g)$
- 500 forward simulations
- Neural posterior: masked autoregressive flow
- 2000 posterior samples drawn

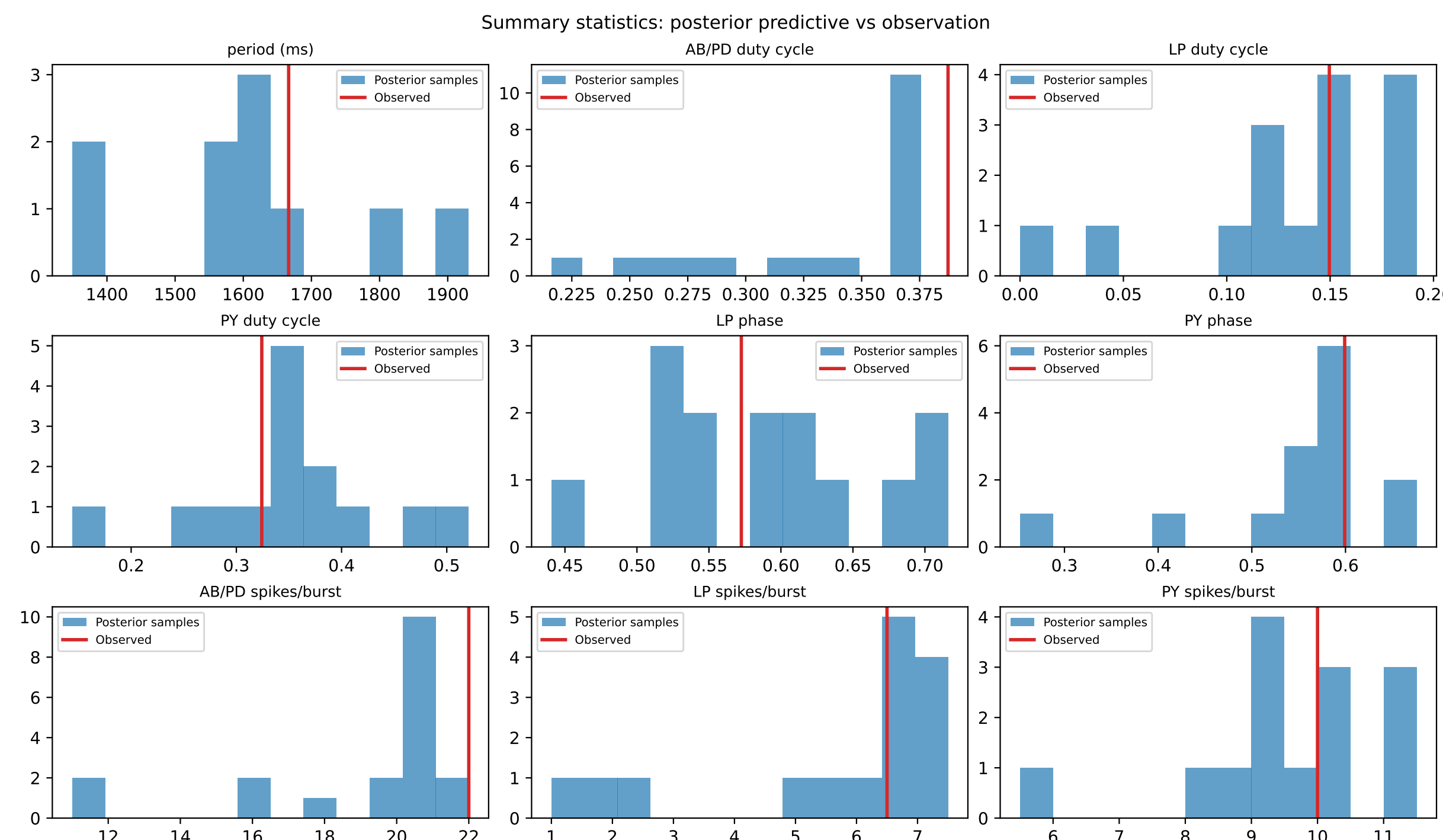
SNPE returns a full **posterior distribution** over all 7 conductances, directly quantifying degeneracy without any additional sampling cost.

Result 1: DE Fit — Observed vs. Simulated



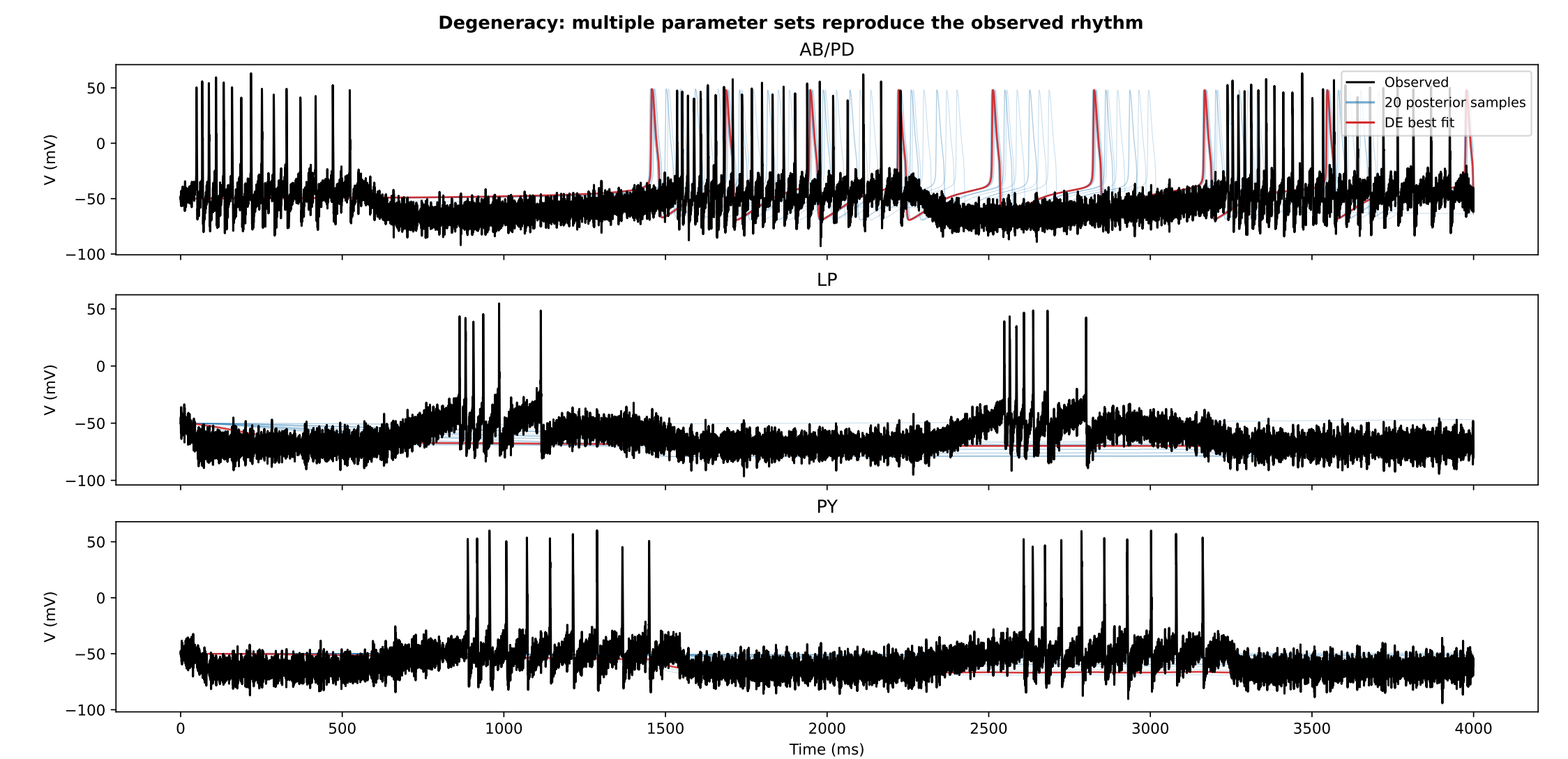
DE best-fit (orange) vs. observed (blue): period, duty cycles, phase offsets, and spike counts matched with **≈7% fractional error** across all 9 features.

Result 2: Posterior Predictive Check



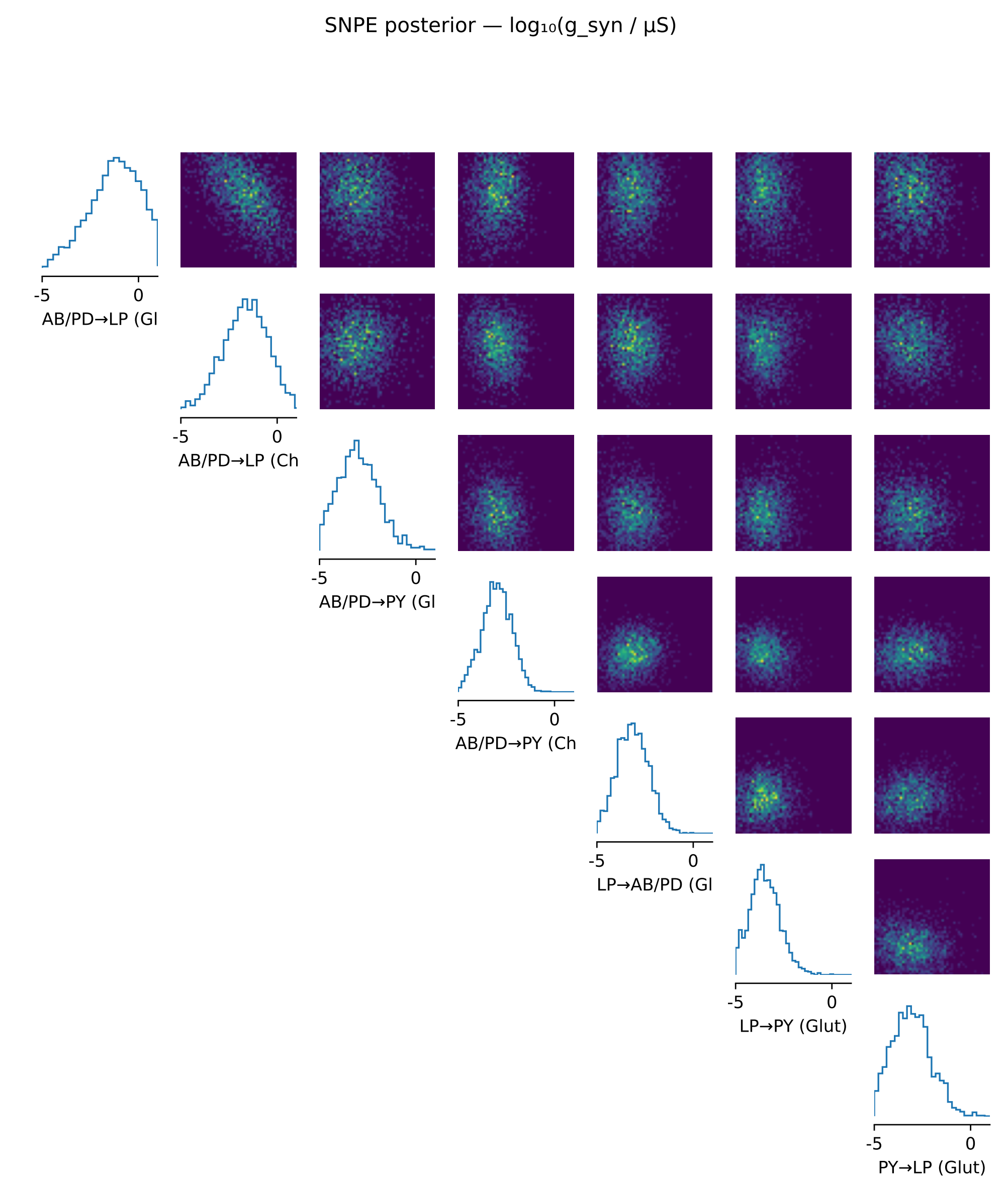
Histograms of summary statistics from 20 posterior samples (blue) vs. the observed target value (red line). Statistics within the posterior predictive distribution confirm that the SNPE posterior captures the data-generating features.

Result 3: Degeneracy — Posterior Trace Overlay



Observed (black), 20 posterior samples (blue), DE best (red). All produce **functionally equivalent rhythms** — different conductances, indistinguishable behaviour.

Result 4: Posterior over Conductances



Pairplot of the SNPE posterior over all 7 log-conductances ($\log_{10} g$). Broad marginals and weak pairwise correlations indicate that the inverse problem is **underdetermined** — a wide range of conductance combinations fits the observation equally well.

Conclusions

- DE recovers a good fit.** Loss of 0.0073 ($\approx 7\%$ fractional error) with a visually convincing trace match.
- The posterior is broad.** SNPE reveals substantial posterior uncertainty, consistent with Prinz et al. (2004).
- Degeneracy is real.** 20 distinct posterior samples all produce valid triphasic rhythms — the recording cannot uniquely identify conductances.
- Phase offsets are most informative.** The 3×-weighted phase features dominate the DE loss and constrain the posterior most tightly.